# When Is Reuse Safe? Regime-Aware Policy Transfer for Recurring Drift in Network Telemetry

Anonymous Submission

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***Abstract*—Network telemetry drifts, and the standard fix, retraining, pays the full cost of rebuilding the model every time. But network regimes recur: a congestion profile, a radio configuration or an operator comes back. We ask whether a policy learned for a regime can be reused when that regime returns, and when doing so is safe. Regime-Aware Policy Transfer (RAPT) stores one ensemble policy per regime, retrieves it on recurrence, and refits it when rolling accuracy shows it no longer fits. On four network streams under a strict prequential protocol and thirty seeds, RAPT uses 64–91% less adaptation CPU than full retraining. Its macro-F1 is lower than full retraining on all four streams, significantly so on all four. The clearest failure is UGR'16, where regime identifiers recur but the feature-to-label mapping does not. Identity-only reuse drops macro-F1 below even the frozen baseline ($0.914$ against $0.969$). We add RAPT-Guard, which scores the stored policy and the prefix policy on the most recent labelled window before every reuse and takes the stored policy only when it wins by a margin. On three of the four streams Guard matches or beats the frozen baseline; on UGR'16 it recovers most of the loss ($0.936$ against $0.969$) while remaining below it. Across all four streams its adaptation CPU is 43–81% below full retraining. Reuse keyed on regime identity alone is therefore unsafe, and a cheap validation gate removes most of the damage but not all of it.**

***Index Terms*—concept drift, network telemetry, online learning, adaptive ensembles, regime-aware learning, computationally efficient adaptation**

## I. Introduction

Network monitoring data does not stay the same for long. Load, radio conditions, subscriber mobility and configuration changes move the distribution, so a classifier fitted on a period of a 5G testbed or an ISP netflow archive will not stay accurate. Retraining once performance drops is the standard answer. It works, but it is paid for again every time, and with high-rate telemetry the total spent rebuilding models can exceed what it costs to serve predictions. Lightweight detection frameworks for constrained devices already exist [1], and many detectors decide *when* to adapt [2]–[8], but they make no use of how the change is structured. Conditions in networks often come back: a weekday congestion profile, a radio configuration or an operator-specific QoS regime may go away and return later. Some of the earliest work in the field was driven by recurring concepts [9], [10], yet the usual practice is still to learn again from scratch, even though the model being thrown away may well be the right one.

The question we ask is whether the discarded information can be kept and used. RAPT keeps a repository of adaptation policies, each indexed by an operating regime. At a regime boundary it looks up whether the incoming regime already has a stored policy. When it does, that ensemble is retrieved at close to zero adaptation cost; when it does not, a new policy is trained and added to the repository. Since a retrieved policy can be wrong if the label relationship in that regime has changed, RAPT also follows its streaming accuracy and refreshes the policy if needed.

If regimes recur with stable label semantics, reuse keeps accuracy and removes most of the adaptation cost. If regimes recur but the feature-to-label relationship has moved, reuse brings back a stale policy and accuracy suffers. We show that this failure is silent, and we make reuse conditional on a cheap validation gate that scores the stored policy against the prefix policy on the most recent labelled window before every reuse. Our contributions are as follows:

- a regime-conditioned policy-transfer mechanism that stores and retrieves ensemble policies by regime;
- RAPT-Guard, a validation gate that turns "has this regime been seen?" into "is the stored policy still better than the one we already have?" at the cost of one extra scoring pass per regime return;
- an evaluation on four network streams under one prequential protocol, thirty seeds, against four established drift detectors and against periodic and event-driven baselines with and without reuse;
- a failure analysis showing that identity-only reuse can fall below the frozen baseline on a stream whose regime identifiers recur but whose label relation does not, and quantifying how much the gate recovers.

## II. Related Work

Drift is split into covariate shift, where $P(X)$ moves and $P(Y \mid X)$ holds, and concept drift, where $P(Y \mid X)$ moves [4], [11]. The field began with recurring concepts [9] and the robustness–flexibility trade-off [10]. Detectors decide *when* to adapt: ADWIN [2], Page-Hinkley [3], DDM and EDD [4], [5], EDMA and EWMA [6], and unlabelled feature tests [7], [8], [12]. Online ensembles decide *how* to adapt [13]–[15]; single-pass learners bound per-example cost [16]; our detectors and incremental learners come from River [17]. Transfer learning moves parameters between related tasks [18], with roots in forgetting [19] and knowledge reuse [20]. None of this work asks whether a model already in hand is good enough to reuse, which is our question.

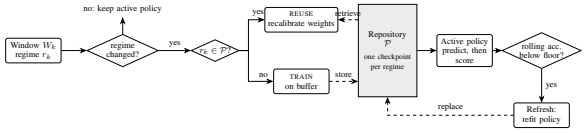



Fig. 1. RAPT overview. At each regime boundary the controller looks up the incoming regime in the repository and either reuses the stored policy or trains and stores a new one. The active policy predicts and is scored as labels arrive. A monitor refits the policy and replaces its checkpoint when rolling accuracy degrades. Solid arrows are control flow, dashed arrows are repository reads and writes.

## III. Problem Formulation

We treat the data as a chronological stream $x_1, x_2, \ldots$, where $x_t \in \mathbb{R}^d$ is a feature vector and $y_t \in \{1, \ldots, C\}$ is its label. The label is unknown until the prediction for $x_t$ has been issued. The stream is divided into fixed-size windows $W_1, \ldots, W_T$, and each window $W_k$ is tagged with a regime identifier $r_k \in \mathcal{R}$. A regime is a semantically defined operating condition, such as a network scenario or a time-of-day block. It is not a cluster label. A *policy* $\pi$ is a fitted model plus its decision rule, and the repository is a mapping $\mathcal{P} : \mathcal{R} \to \pi$ that is filled in as new regimes show up. Whenever the stream moves from $r_{k-1}$ to $r_k$, the controller picks an action $a_k \in \{\text{REUSE}, \text{TRAIN}\}$. It reuses $\mathcal{P}(r_k)$ when that exists, and trains a new policy on the recent buffer when it does not. Let $M(\pi, W)$ be a performance measure, which is macro-F1 in this paper, and let $c(\pi)$ be the cost of producing $\pi$ in adaptation CPU time. The objective is to hold $M(\cdot, W_k)$ near the level a freshly trained policy would reach while making $\sum_k c(a_k)$ as small as possible.

Reuse rests on an assumption, namely that $P(Y \mid X, r_k)$ has not changed since the policy was stored. If the regime identifier returns after the conditional relationship has shifted, the stale policy gets applied and nothing flags it. The hard part is separating a recurring regime from a recurring label relationship.

Every adaptive model draws its training data from one *class-anchored buffer*: the most recent observations up to a capacity, into which a stratified slice of the training prefix (fifteen samples per class) is always mixed, so a refit on a single-class window cannot collapse onto that class. The RAPT-Guard validation lookback comes from the same buffer.

## IV. Proposed Method: RAPT

### A. Architecture and Repository

RAPT sits between the stream and the model and makes one decision at each regime boundary. The active policy predicts each window, and the window is scored when its labels arrive. A change of policy happens only at a regime transition or when monitoring reports degradation.

We represent a regime with its semantic identifier and do not learn a clustering. What we give up is that the mechanism requires a usable regime signal; see Section VIII. The repository holds one checkpoint per regime, made up of the fitted ensemble, its soft-voting weights, and timestamps for creation and last access. Checkpoints are immutable.

### B. Matching and Transfer

Matching uses the regime identifier: on a transition to regime $r$, the controller tests whether $r \in \mathcal{P}$ and retrieves the checkpoint if it does. When reuse goes ahead, the retrieved ensemble becomes the active policy, and its soft-voting weights are recalibrated on the most recent buffered window (at most the last 50 observations, skipped below 20). If no checkpoint exists, a new policy is trained from the class-anchored buffer and stored. All adaptive models use that buffer in the same way.

### C. Monitoring, Refresh, and Fallback

We use two monitoring rules, and how they differ is one of our findings. The *absolute* rule triggers when rolling accuracy goes under a fixed floor. The *relative* rule triggers when accuracy has dropped by more than a fixed margin from a decayed reference built from that policy's own earlier accuracy. A relative rule cannot catch a policy that was never good, because the reference decays toward the poor level and the gap it looks for never appears.

A refresh rebuilds the active policy from the recent buffer and replaces its checkpoint. In the default refresh the 100-tree ensemble is refit on a 1000-observation buffer; the cost-aware refresh has the same trigger but refits 20 trees on a 300-observation buffer.

*RAPT-Guard* replaces the indirect degradation monitor with a direct test at the moment of reuse. When the stream returns to a regime $r$ that already has a checkpoint, the controller scores the stored ensemble and the prefix-trained ensemble on the most recent $L$ labelled buffer rows, and reuses the stored policy only if its accuracy is at least that of the prefix policy plus a margin $m$; otherwise it trains a fresh policy on the buffer and stores it as the new checkpoint. The gate asks whether reuse is still better than the fallback we already hold, rather than whether a policy that was never good has degraded. The lookback is at most one window, so the gate adds one scoring pass per regime return. We select $(L, m)$ on the development split subject to the reuse rate staying above $0.30$, so the gate cannot collapse onto "always train". RAPT-Enhanced adds buffer-blended novelty refitting (a 1500-observation buffer instead of 500) and selective parity refitting (refit a reused policy once its rolling accuracy falls under a parity threshold for several windows, which reacts only after the stale policy has predicted, unlike the prospective gate). When no checkpoint exists, or the similarity gate turns it down, a fresh policy is trained and stored under the incoming regime identifier.

## V. Experimental Methodology

### A. Datasets

Four datasets are summarized in Tab. I; all are real network traces. *5G Campus* is a testbed QoS stream with delay metrics labelled GOOD, DEGRADED or BAD from the 90th-percentile delay of the next window; 1,790 usable samples give 179 windows of 10, each of the three scenarios recurs fourteen times. *UGR'16* is aggregated ISP netflow with 134

features. Because the calibration split contains almost no labelled attack traffic, we use the first thirty days of labelled data, one sample per minute (43,200 samples); time-of-day crossed with weekday/weekend gives twelve regimes and 168 recurrences. *NordicDat* is a cross-border LTE/5G predictive-QoS trace [21] whose regimes are operators; it is strongly imbalanced (the smallest class is $15\%$ of the stream) and is our hardest stream. *5G NR* is a simulated end-to-end latency stream [22] of 499 windows of 500 packets across four regimes (ABCDACBDAB), labelled with the same next-window rule as Campus. Campus and NordicDat share this three-class next-window delay rule; UGR'16 is binary.

TABLE I
DATASETS USED IN EVALUATIONS. A WINDOW IS THE ADAPTATION STEP; 5G NR IS ONE ROW PER WINDOW. RECURRENCES ARE REVISITS OF AN ALREADY-SEEN REGIME.

| **Property** | **Campus** | **UGR'16** | **Nordic** | **5G NR** |
|---|---|---|---|---|
| Total samples | 1,799 | 43,200 | 91,455 | 499 |
| Usable samples | 1,790 | 43,200 | 91,000 | 499 |
| Features | 19 | 134 | 16 | 12 |
| Classes | 3 | 2 | 3 | 3 |
| Windows | 179 | 180 | 182 | 499 |
| Samples/window | 10 | 240 | 500 | 1 |
| Regimes | 3 | 12 | 3 | 4 |
| Initial-train windows | 36 | 36 | 36 | 100 |

### *B. Preprocessing, Ordering, and Leakage Control*

All four streams undergo the exact same processing. First, all four streams undergo sorting by row according to timestamp. Then, all identifier and constant columns are dropped. Finally, only those features that are numeric are retained. Any infinite values present within a stream are treated as missing. Any rows that lack a target value are deleted. Any remaining gaps are then filled using medians computed solely from the training prefix. Target thresholds and any scale-related statistics also derive from that prefix. None of the streams are shuffled. The prefix is the first twenty percent of windows: thirty-six for the three cross-dataset streams and ninety-nine for 5G NR. That prefix is completely excluded from evaluation, and that is the sole source of target thresholds. The order of operations is predict $\rightarrow$ evaluate $\rightarrow$ adapt for every window prior to its labels being used for adaptation. Regime annotation occurs directly from reading the stream. Therefore, no derivation of annotation can occur based upon the target.

### *C. Models And Detectors*

Each of the five primary models utilizes exactly the same base architecture and the same retraining buffer. Therefore, any variation between them is due to their respective adaptation mechanisms only. The base is an ensemble of a random forest classifier and an extra-trees classifier, both having fifty trees of depth seven. Frozen is trained once on the prefix and receives no subsequent updates. Event-Driven updates when window accuracy falls below a fixed threshold. Full Retraining rebuilds the model at every regime transition. RAPT, RAPT-Enhanced and RAPT-Guard are the controllers of Section IV. We also report two baselines that isolate reuse from the trigger: Frozen+Periodic refits on a fixed five-window schedule, and Event-Driven+Reuse uses the event trigger but retrieves a stored policy when the current regime has one. RAPT-Anchor is an oracle upper bound on the gate: each window it predicts with whichever of the active policy and the prefix policy scored better on the labelled lookback, so its inference cost is doubled and charged as such.

ADWIN, Page-Hinkley, EDD and EDMA operate within the same retraining harness, with identical model, buffer, capacity and minimum-retrain-interval. They differ only by how they detect triggers. Following the finding that the historical default parameters left ADWIN and Page-Hinkley silent, we select their operating points on the development split by a fire-rate criterion (median 5–20 events per stream, with the minimum stream event count maximised) rather than against the evaluation streams. The frozen values are ADWIN $\delta = 0.2$; Page-Hinkley $\alpha = 0.9999$, $\delta = 0.005$, threshold $= 0.002$, min-instances $= 10$; EDD warning $= 2$, drift $= 3$; EDMA $\alpha = 0.3$, $k = 1.5$. All four now consume the per-sample 0/1 error stream, which is the input their original definitions assume. Since none of our streams carry ground-truth drift boundary annotations we report detected events, adaptation cost and predictive performance, not detector precision or recall.

### *D. Metrics And Statistics*

Performance is evaluated through macro-F1, accuracy, precision and recall. Costs are measured two ways. *Adaptation CPU* is the CPU time devoted to adaptation routines (refits, checkpoint writes, and the gate's scoring passes) and *total CPU* is adaptation plus inference, so a method that infers twice is charged twice. A reduction in adaptation time does not translate proportionally into total runtime, which is why we report both; the total-CPU column is what a deployment actually pays. Each experiment uses thirty seeds (42–71) and we report mean and standard deviation over seeds. Comparisons use the Wilcoxon signed-rank test (paired by window, with Holm correction across the comparison family) and Cohen's $d_z$ on the per-seed paired differences, plus two one-sided tests (TOST) for equivalence at a $\pm 0.02$ macro-F1 margin. With thirty seeds the seed-level tests are no longer power-limited; the minimum attainable two-sided $p$-value is $2^{-30}$.

### *E. Predictive Performance*

Table II holds the primary comparison. On 5G Campus and 5G NR the models cluster within $0.03$ macro-F1, while UGR'16 and NordicDat separate them sharply. On UGR'16 the frozen baseline is the best model, which is the signature of stale-policy reuse: the regimes recur, but the label relation does not, so retrieving the stored policy is worse than never adapting at all.

On 5G Campus every adaptive model is close to saturation. Full Retraining has the best macro-F1 ($0.9836$), while RAPT reaches $0.9385$; the gap of $0.0451$ is significant ($p = 1.4 \times$

TABLE II

PRIMARY COMPARISON, MEAN OVER 30 SEEDS. BOLD MARKS THE BEST MACRO-F1 PER STREAM; CPU IN SECONDS, RETRAINS AND REUSES ARE COUNTS. RAPT-ANCHOR'S TOTAL CPU INCLUDES ITS DOUBLED INFERENCE.

| Dataset | Model | Macro-F1 | Adapt CPU | Total CPU | Retr. | Reuse |
|---|---|---|---|---|---|---|
| Campus | Frozen | 0.9371 | 0.000 | 1.185 | 0 | 0 |
| | Event-Driven | 0.9638 | 0.166 | 1.358 | 1 | 0 |
| | Full Retraining | **0.9836** | 2.192 | 3.375 | 14 | 0 |
| | RAPT | 0.9385 | 0.281 | 1.452 | 2 | 12 |
| | RAPT-Enhanced | 0.9385 | 0.279 | 1.431 | 2 | 12 |
| | RAPT-Guard | 0.9391 | 0.499 | 1.675 | 2 | 12 |
| | RAPT-Anchor | 0.9385 | 0.282 | 3.893 | 2 | 12 |
| UGR'16 | Frozen | **0.9693** | 0.000 | 1.037 | 0 | 0 |
| | Event-Driven | 0.8978 | 0.835 | 1.942 | 4 | 0 |
| | Full Retraining | 0.9511 | 29.771 | 30.872 | 144 | 0 |
| | RAPT | 0.9137 | 2.571 | 3.640 | 11 | 133 |
| | RAPT-Enhanced | 0.9137 | 2.595 | 3.669 | 11 | 133 |
| | RAPT-Guard | 0.9357 | 5.743 | 6.810 | 17 | 127 |
| | RAPT-Anchor | 0.9247 | 2.595 | 5.683 | 11 | 133 |
| Nordic | Frozen | 0.2851 | 0.000 | 1.396 | 0 | 0 |
| | Event-Driven | 0.2739 | 0.346 | 1.706 | 2 | 0 |
| | Full Retraining | **0.4376** | 3.310 | 4.644 | 19 | 0 |
| | RAPT | 0.3845 | 0.344 | 1.679 | 2 | 16 |
| | RAPT-Enhanced | 0.3775 | 3.686 | 5.061 | 21 | 16 |
| | RAPT-Guard | 0.3793 | 1.154 | 2.537 | 5 | 13 |
| | RAPT-Anchor | 0.3838 | 0.351 | 4.130 | 2 | 16 |
| 5G NR | Frozen | 0.9110 | 0.000 | 2.709 | 0 | 0 |
| | Event-Driven | 0.9081 | 1.312 | 3.984 | 11 | 0 |
| | Full Retraining | 0.9121 | 0.981 | 3.769 | 8 | 0 |
| | RAPT | 0.8969 | 0.350 | 3.092 | 3 | 5 |
| | RAPT-Enhanced | 0.9045 | 0.963 | 3.677 | 8 | 5 |
| | RAPT-Guard | **0.9148** | 0.558 | 3.221 | 4 | 4 |
| | RAPT-Anchor | 0.9042 | 0.349 | 8.851 | 3 | 5 |

$10^{-91}$, $d_z = -12.5$). RAPT-Guard edges it up to 0.9391, still 0.0445 below Full Retraining and still significant. Here the gate is not what is missing: reuse on this stream brings back a policy that is simply less accurate than one trained on the current buffer, and the gate correctly prefers the prefix model but the prefix model is also weaker than Full Retraining.

UGR'16 is where the assumption of the mechanism fails, and it teaches us the most. Frozen has the highest macro-F1 (0.9693). Identity-only reuse is the worst of the eight primary models: RAPT reaches only 0.9137, which is 0.0556 below Frozen ($p = 1.0 \times 10^{-68}$, $d_z = -3.6$) and 0.0375 below Full Retraining, despite its 133 reuses. The twelve regimes recur and their identifiers stay valid across the 30-day block, but the netflow-feature-to-attack-label mapping does not stay put. RAPT-Guard, which validates before reusing, recovers to 0.9357. That is 0.0220 above base RAPT and 0.0336 below Frozen; both gaps are significant ($p = 9.0 \times 10^{-7}$ and $p = 1.3 \times 10^{-32}$). So the gate repairs most, but not all, of the damage, and on this stream it does not beat the frozen baseline. RAPT-Enhanced's parity refit reaches 0.9137, the same as base RAPT here, because a policy that is poor from the first reused window is refit late.

NordicDat is hard to predict and strongly imbalanced. Frozen scores 0.2851, adaptation helps, and Full Retraining gets 0.4376 against 0.3845 for RAPT, a deficit of 0.0531 ($p = 6.1 \times 10^{-4}$, $d_z = -1.1$). RAPT-Guard scores 0.3793, 0.0583 below Full Retraining. Unlike UGR'16, identity-only reuse here is not catastrophic: RAPT stays above Frozen ($p = 8.4 \times 10^{-73}$), so reuse is safe on this stream even though

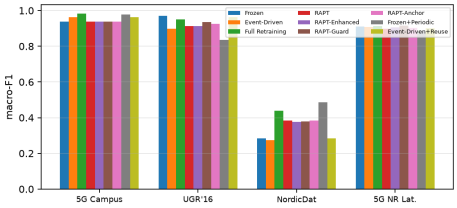



Fig. 2. Macro-F1 per method and stream, mean over 30 seeds (Table II).

it is not accurate. On 5G NR the models are within 0.03 of one another and RAPT-Guard has the best macro-F1 of all methods (0.9148), though its edge over Full Retraining (0.9121) is not significant ($p = 0.15$). Taken together, the predictive results give no basis for saying that RAPT improves accuracy. Full Retraining has the best macro-F1 on NordicDat, the frozen baseline on UGR'16, and a detector baseline on Campus; on 5G NR the best nominal model is RAPT-Guard. Base RAPT sits below Full Retraining on all four streams. The case for RAPT is cost, and the case for RAPT-Guard is that it makes reuse safe enough to use where identity-only reuse is not.

### *F. Computational Cost and Policy Reuse*

Table II lists both adaptation and total CPU; the total column is what a deployment pays, because it also charges inference. Relative to Full Retraining the adaptation CPU falls on every stream: 2.192 → 0.281 s on 5G Campus (−87.2%), 29.771 → 2.571 s on UGR'16 (−91.4%), 3.310 → 0.344 s on NordicDat (−89.6%), and 0.981 → 0.350 s on 5G NR (−64.3%). Full

TABLE III

RAPT-GUARD DECISIONS AT EVERY REGIME RETURN, THREE SEEDS, WITH A TWO-POINT TOLERANCE. BAD ALLOW COUNTS A REUSE THAT LOST MORE THAN TWO MACRO-F1 POINTS TO THE PREFIX POLICY ON THE INCOMING WINDOW.

| Stream | Returns | Allow | Block | Bad allow | Bad block | Bad-allow rate |
|---|---|---|---|---|---|---|
| Campus | 36 | 36 | 0 | 0 | 0 | |
| UGR'16 | 399 | 385 | 14 | 6 | 0 | |
| Nordic | 48 | 39 | 9 | 3 | 3 | |
| 5G NR | 15 | 11 | 4 | 0 | 0 | |

TABLE IV

HISTORICAL DETECTORS AND THE REUSE METHODS AT THE FROZEN OPERATING POINTS, MEAN OVER 30 SEEDS. EACH CELL IS MACRO-F1 WITH RETRAIN EVENTS PER SEED IN PARENTHESES.

| Method | Campus | UGR'16 | Nordic | 5G NR |
|---|---|---|---|---|
| ADWIN | 0.937 (0.0) | 0.950 (8.7) | 0.328 (30.5) | 0.911 (0.1) |
| Page-Hinkley | 0.984 (6.0) | 0.960 (13.3) | 0.468 (34.1) | 0.905 (19.6) |
| EDDM | 0.984 (6.0) | 0.960 (13.3) | 0.433 (32.1) | 0.910 (7.4) |
| EDMA | 0.984 (6.0) | 0.960 (13.3) | 0.378 (29.6) | 0.907 (9.0) |
| Event-Driven | 0.964 (1.0) | 0.898 (3.6) | 0.274 (2.0) | 0.908 (10.9) |
| RAPT | 0.939 (2.0) | 0.914 (11.0) | 0.385 (2.0) | 0.897 (3.0) |

Retraining retrains at every regime transition (14, 144, 19 and 8 times); RAPT retrains 2, 11, 2 and 3 times and retrieves a stored policy 12, 133, 16 and 5 times. The gate costs more than identity-only reuse but stays far below Full Retraining: RAPT-Guard's adaptation CPU is 77.2%, 80.7%, 65.1% and 43.1% below Full Retraining on the four streams. The doubled inference of RAPT-Anchor is visible in the total-CPU column: on 5G NR it totals 8.851 s, more than Full Retraining's 3.769 s, even though its adaptation CPU is the lowest of the family. This is why the fair cost column matters: a method can win on adaptation and lose on total.

### G. Efficiency Ablation

On 5G Campus (five seeds) the mechanism ladder is: RAPT-Full $0.9587$ at $0.4283$ s; RAPT-Evidence fires *zero* refreshes; RAPT-Floor fires 7.6 refreshes, $0.9797$ at $0.5865$ s; RAPT-Cheap $0.9851 \pm 0.0029$ at $0.8477$ s against Full Retraining $0.9829 \pm 0.0023$ at $1.4063$ s ($\Delta = +0.0022$, $p = 0.3125$, 95% interval $[-0.0036, +0.0080]$ includes zero); RAPT-Incremental $0.9549$ at $0.5813$ s is worse on both axes and is dropped. These use the legacy five-seed harness and are power-limited.

### H. Validation-Gate Accuracy

RAPT-Guard turns every regime return into a decision, so we can audit it. For each return we compare the gate's choice with the outcome on the incoming window, using a two-point tolerance: a reuse is harmful only if the stored policy scores more than two points below the prefix policy, and a block is a miss only if reuse would have been more than two points better. Table III reports the counts. The gate almost never allows a harmful reuse: 0 of 36 on Campus, 6 of 385 on UGR'16 (1.6%), 3 of 39 on NordicDat (7.7%), 0 of 11 on 5G NR; it is slightly conservative on NordicDat, blocking 3 reuses that would have helped. The zero margin was chosen on the development split to keep the gate a gate (allow rate above 0.30) while maximising development macro-F1; a positive margin blocks almost everything and a negative margin allows everything.

### I. Comparison to Historical Detectors

Table IV reports the four historical detectors at the frozen development operating points; they no longer reduce to the frozen baseline. Page-Hinkley is strongest on Campus and NordicDat (0.984, 0.468) and matches EDD and EDMA on

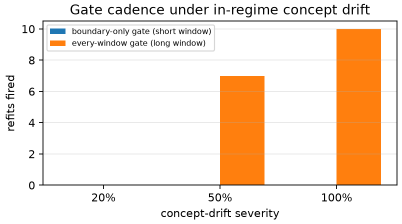



Fig. 3. Short- versus long-window gate under the concept-drift stress panel.

UGR'16 (0.960), but fires 6–34 times per stream and pays the most adaptation CPU (2.47–5.44 s). ADWIN fires 8.7 times on UGR'16 and 30 on NordicDat yet has the weakest NordicDat macro-F1 (0.328): firing more is not adapting well. RAPT fires 2–11 times and RAPT-Guard 4–17 times, well below the detectors at a fraction of their cost. On Campus the detectors reach Full-Retraining accuracy at lower cost, so RAPT is not the accuracy leader there; on UGR'16 the frozen baseline still wins.

### J. Gate Cadence: Short versus Long Window

The gate can be evaluated on the incoming window only (short, boundary-only lookback) or on every window while a reused policy is active (long, continuous lookback). Under a synthetic concept-drift stress panel that reverses the rank order of the top predictive features inside a regime, the boundary-only cadence never fires a refit, because the drift is in-regime and there is no boundary to key on; the every-window cadence fires 7–10 refits at the two higher severities. The macro-F1 effect is small where the window is still predictable ($0.8902 \rightarrow 0.8941$ at severity 0.5, no change at 0.2 and 1.0). The long-window cadence detects in-regime drift the short-window cadence cannot see at all, but neither moves macro-F1 much once the window is near-single-class, so we claim a detection mechanism and no large accuracy effect.

## VI. DISCUSSION

The mechanism is worth deploying when there is recurrence. The cost saving appears wherever reuse occurs, but the accuracy penalty is concentrated on UGR'16, where regimes recur and their label relationship does not. Reusing on the regime identifier alone is unsafe when label semantics can drift independently of it, and even a direct validation gate only repairs most of the damage. The practical rule is narrow: reuse

when regimes recur *and* validate the stored policy against the prefix fallback at every return; if validation finds no stored policy that beats the fallback, do not reuse.

## VII. Independent Re-validation

A harness that re-implements the protocol from the raw per-window records reproduces Table II on every predictive metric and regenerates the statistics, gate counts and detector events; a re-run reproduced every metric exactly but shifted adaptation CPU by up to $0.14\,\mathrm{s}$, so the ordering is stable and the seconds indicative. Retrieval keys do not track label semantics: an oracle retraining on the previous visit of the *same* regime reaches only $0.5926$ macro-F1 on UGR'16 against $0.9595$ for Full Retraining, but $0.9905$ on Campus, exactly where identity-only reuse collapses. The added controls separate the trigger from reuse: Frozen+Periodic is the best model on NordicDat ($0.4872$, above Full Retraining's $0.4376$), and on UGR'16 Event-Driven+Reuse recovers to $0.9639$, above Event-Driven ($0.8978$) and base RAPT ($0.9137$) though still below Frozen ($0.9693$). A label-delay variant costs RAPT $0.13$ macro-F1 on NordicDat; the headline numbers assume labels arrive with their window.

## VIII. Limitations and Threats to Validity

The seed-level tests use thirty seeds, so they are not power-limited, but the mechanism ablation in Section V-G uses five seeds and its smallest two-sided $p$-value is $0.0625$. Window-level tests assume windows from one stream are independent, which inflates the effective sample size. The mechanism relies on regime recurrence, which we measure and do not assume: a stream with no recurrence never exercises the repository, and a novel regime falls back to retraining with no advantage over the baseline. The RAPT-Floor threshold and the 20-tree cheap refit were calibrated for this ensemble and would need revisiting where the pre/post-drift gap is larger. NordicDat stays hard (best macro-F1 $0.4872$) and we flag it as a hard stream rather than a success. Finally, the repository is never pruned, and regime similarity is treated as meaningful, which may fail if the regime signal is noisy.

## IX. Conclusion

Retraining is the standard response to drift, but when conditions return, part of what was paid for is already in a previously trained model. RAPT keeps one policy per regime and retrieves it on recurrence, training anew only when no suitable policy exists. Across four network streams adaptation CPU was 64–91% below full retraining, while macro-F1 was lower on all four. Where regimes recurred with changed label relations, identity-only reuse dropped below even the frozen baseline; RAPT-Guard, which validates the stored policy against the prefix fallback before every reuse, recovered most of that loss but on UGR'16 still did not beat the frozen baseline. Policy reuse is worth deploying when regimes recur and the stored policy is validated at every return, and the natural next step is to test whether a retrieved regime still means what it did when its policy was stored.

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